

AREG

Automotive Radar Echo Generator

Getting Started

R&S®AREG800A



1179360902
Version 06

ROHDE & SCHWARZ
Make ideas real



This document describes the AREG, stock no. 1437.4400.02.

© 2024 Rohde & Schwarz

Muehldorfstr. 15, 81671 Muenchen, Germany

Phone: +49 89 41 29 - 0

Email: info@rohde-schwarz.com

Internet: www.rohde-schwarz.com

Subject to change – data without tolerance limits is not binding.

R&S® is a registered trademark of Rohde & Schwarz GmbH & Co. KG.

All other trademarks are the properties of their respective owners.

1179.3609.02 | Version 06 | AREG

Throughout this document, R&S® is indicated as R&S.

Contents

1 Safety and regulatory information.....	7
1.1 Safety instructions.....	7
1.2 Labels on AREG.....	11
1.3 Warning messages in the documentation.....	11
1.4 Where to find key documents on Rohde & Schwarz.....	12
1.5 Korea certification class A.....	12
2 Documentation overview.....	13
2.1 Getting started manual.....	13
2.2 User manuals and help.....	13
2.3 Service manual.....	13
2.4 Instrument security procedures.....	14
2.5 Printed safety instructions.....	14
2.6 Specifications and product brochures.....	14
2.7 Calibration certificate.....	14
2.8 Release notes and open source acknowledgment.....	15
2.9 Application notes, application cards, white papers, etc.....	15
2.10 Videos.....	15
3 Key features.....	16
4 Preparing for use.....	17
4.1 Lifting and carrying.....	17
4.2 Unpacking and checking.....	17
4.3 Choosing the operating site.....	18
4.4 Setting up the AREG.....	18
4.5 Considerations for test setup.....	20

4.6 Connecting to power.....	21
4.7 Connecting to LAN.....	22
4.8 Connecting USB devices.....	23
4.9 Connecting to RF coaxial connectors.....	24
4.10 Connecting the R&S QAT100.....	26
4.11 Connecting to Ref In/Ref Out.....	27
4.12 Connecting to Digital IO DD.....	27
4.13 Connecting to Aux IF In/Aux IF Out.....	28
4.14 Connecting a TRX frontend.....	28
4.15 Switching on or off.....	31
5 Instrument tour.....	33
5.1 Front panel tour.....	33
5.2 Rear panel tour.....	39
6 Trying out the instrument.....	46
6.1 Configuring static operation setups.....	46
6.2 Saving and recalling settings.....	48
7 Operating the instrument.....	51
7.1 Means of manual interaction.....	52
7.2 Understanding the display information.....	52
7.3 Accessing the functionality.....	57
7.4 Entering data.....	58
7.5 Undo and redo actions.....	60
7.6 Getting information and help.....	60
7.7 Remote operation over web UI.....	62
8 Contacting customer support.....	64
Index.....	65

1 Safety and regulatory information

The product documentation helps you use the product safely and efficiently. Follow the instructions provided here and in the following sections.

Intended use

The product generates radio frequency (RF) signals for the development, production and verification of electronic components, modules or devices. The product is intended for industrial use, for example for production and conformance testing, maintenance and engineering laboratories.

Use the product only for its designated purpose. Any other use is considered improper use. Observe the operating conditions and performance limits stated in the specifications document.

Target audience

This document targets at all users, including technicians, operators, administrators and maintenance personnel. The required skills and experience of the users depend on the test setup and application of the product.

Where do I find safety information?

Safety information is part of the product documentation. It warns you of potential dangers and gives instructions on how to prevent personal injury or damage caused by dangerous situations. Safety information is provided as follows:

- In [Chapter 1.1, "Safety instructions"](#), on page 7. The same information is provided in many languages in printed format. The printed "Safety Instructions" for "Mains-Powered Products, Heavy" (document number 1171.1788.99) are delivered with the product.
- Throughout the documentation, safety instructions are provided when you need to take care during setup or operation.

1.1 Safety instructions

Products from the Rohde & Schwarz group of companies are manufactured according to the highest technical standards. To use the products safely, follow

the instructions provided here and in the product documentation. Keep the product documentation nearby and offer it to other users.

Use the product only for its intended use and within its performance limits. Intended use and limits are described in the product documentation such as the specifications document, manuals and the printed "Safety Instructions" document. If you are unsure about the appropriate use, contact Rohde & Schwarz customer support.

Using the product requires specialists or specially trained personnel. These users also need sound knowledge of at least one of the languages in which the user interfaces and the product documentation are available.

Reconfigure or adjust the product only as described in the product documentation or the specifications document. Any other modifications can affect safety and are not permitted.

Never open the casing of the product. Only service personnel authorized by Rohde & Schwarz are allowed to repair the product. If any part of the product is damaged or broken, stop using the product. Contact Rohde & Schwarz customer support at <https://www.rohde-schwarz.com/support>.

Lifting and carrying the product

The product is heavy. Do not move or carry the product by yourself. A single person can only carry a maximum of 18 kg safely depending on age, gender and physical condition. Look up the maximum weight in the specifications document. Use the product handles to move or carry the product. Do not lift by the accessories mounted on the product. Accessories are not designed to carry the weight of the product.

To move the product safely, you can use lifting or transporting equipment such as lift trucks and forklifts. Follow the instructions provided by the equipment manufacturer.

Choosing the operating site

Only use the product indoors. The product casing is not waterproof. Water that enters can electrically connect the casing with live parts, which can lead to electric shock, serious personal injury or death if you touch the casing.

If Rohde & Schwarz provides accessories designed for outdoor use of your product, e.g. a protective cover, you can use the product outdoors.

You can operate the product up to an altitude of 2000 m above sea level. If a higher altitude is permissible, the value is provided in the specifications document. The product is suitable for pollution degree 2 environments where nonconductive contamination can occur. For more information on environmental conditions such as ambient temperature and humidity, see the specifications document.

Setting up the product

Always place the product on a stable, flat and level surface with the bottom of the product facing down. If the product is designed for different positions, secure the product so that it cannot fall over.

If the product has foldable feet, always fold the feet completely in or out to ensure stability. The feet can collapse if they are not folded out completely or if the product is moved without lifting it. The foldable feet are designed to carry the weight of the product, but not an extra load.

If stacking is possible, keep in mind that a stack of products can fall over and cause injury.

If you mount products in a rack, ensure that the rack has sufficient load capacity and stability. Observe the specifications of the rack manufacturer. Always install the products from the bottom shelf to the top shelf so that the rack stands securely. Secure the product so that it cannot fall off the rack.

Connecting the product

Before connecting the interfaces and measuring inputs of the product to other products or electrical circuits, make sure that the other products or electrical circuits provide special protection against electric shock. This protection principle is referred to as SELV (safety extra-low voltage) and is based on a low voltage level and increased insulation. Exceptions are indicated by a measurement category on the product and given in the specifications document.

Connecting to power

The product is an overvoltage category II product. Connect the product to a fixed installation used to supply energy-consuming equipment such as household appliances and similar loads. Keep in mind that electrically powered products have risks, such as electric shock, fire, personal injury or even death. Replace parts that are relevant to safety only by original parts, e.g. power cables or fuses.

Take the following measures for your safety:

- Before switching on the product, ensure that the voltage and frequency indicated on the product match the available power source. If the power adapter does not adjust automatically, set the correct value and check the rating of the fuse.
- Only use the power cable delivered with the product. It complies with country-specific safety requirements. Only insert the plug into an outlet with protective conductor terminal.
- Only use intact cables and route them carefully so that they cannot be damaged. Check the power cables regularly to ensure that they are undamaged. Also ensure that nobody can trip over loose cables.
- Only connect the product to a power source with a fuse protection of maximum 20 A.
- Ensure that you can disconnect the product from the power source at any time. Pull the power plug to disconnect the product. The power plug must be easily accessible. If the product is integrated into a system that does not meet these requirements, provide an easily accessible circuit breaker at the system level.

Cleaning the product

Use a dry, lint-free cloth to clean the product. When cleaning, keep in mind that the casing is not waterproof. Do not use liquid cleaning agents.

Meaning of safety labels

Safety labels on the product warn against potential hazards.

	Potential hazard Read the product documentation to avoid personal injury or product damage.
	Heavy product Be careful when lifting, moving or carrying the product. Carrying the product requires a sufficient number of persons or transport equipment.
	Electrical hazard Indicates live parts. Risk of electric shock, fire, personal injury or even death.

Warning messages in the documentation

	Hot surface Do not touch. Risk of skin burns. Risk of fire.
	Protective conductor terminal Connect this terminal to a grounded external conductor or to protective ground. This connection protects you against electric shock if an electric problem occurs.

1.2 Labels on AREG

Labels on the casing inform about:

- Personal safety, see "[Connecting to power](#)" on page 9.
- Product and environment safety, see [Table 1-1](#).
- Identification of the product, see the serial number on the rear panel.

Table 1-1: Labels regarding AREG and environment safety

	Labeling in line with EN 50419 for disposal of electrical and electronic equipment after the product has come to the end of its service life. For more information, see the product user manual, chapter "Disposal".
---	--

1.3 Warning messages in the documentation

A warning message points out a risk or danger that you need to be aware of. The signal word indicates the severity of the safety hazard and how likely it will occur if you do not follow the safety precautions.

WARNING

Potentially hazardous situation. Could result in death or serious injury if not avoided.

CAUTION

Potentially hazardous situation. Could result in minor or moderate injury if not avoided.

NOTICE

Potential risks of damage. Could result in damage to the supported product or to other property.

1.4 Where to find key documents on Rohde & Schwarz

Certificates issued to Rohde & Schwarz that are relevant for your country are provided at www.rohde-schwarz.com/key-documents, e.g. concerning:

- Quality management
- Environmental management
- Information security management
- Accreditations

1.5 Korea certification class A



이 기기는 업무용(A급) 전자파 적합기기로서 판매자 또는 사용자는 이 점을 주의하시기 바라며, 가정외의 지역에서 사용하는 것을 목적으로 합니다.

2 Documentation overview

This section provides an overview of the AREG user documentation. Unless specified otherwise, you find the documents at:

www.rohde-schwarz.com/manual/areg800a

2.1 Getting started manual

Introduces the AREG and describes how to set up and start working with the product. Includes basic operations, typical measurement examples, and general information, e.g. safety instructions, etc. A printed version is delivered with the instrument.

2.2 User manuals and help

Contains the description of all instrument modes and functions. It also provides an introduction to remote control, a complete description of the remote control commands with programming examples, and information on maintenance, instrument interfaces and error messages. Includes the contents of the getting started manual.

The contents of the user manuals are available as help in the AREG. The help offers quick, context-sensitive access to the complete information.

All user manuals are also available for download or for immediate display on the internet.

2.3 Service manual

Describes the performance test for checking compliance with rated specifications, firmware update, troubleshooting, adjustments, installing options and maintenance.

The service manual is available for registered users on the global Rohde & Schwarz information system (GLORIS):

<https://gloris.rohde-schwarz.com>

2.4 Instrument security procedures

Deals with security issues when working with the AREG in secure areas. It is available for download on the internet.

2.5 Printed safety instructions

Provides safety information in many languages. The printed document is delivered with the product.

2.6 Specifications and product brochures

The specifications document, also known as the data sheet, contains the technical specifications of the AREG. It also lists the firmware applications and their order numbers, and optional accessories.

The brochure provides an overview of the instrument and deals with the specific characteristics.

See www.rohde-schwarz.com/brochure-datasheet/areg800a

2.7 Calibration certificate

The document is available on <https://gloris.rohde-schwarz.com/calcert>. You need the device ID of your instrument, which you can find on a label on the rear panel.

2.8 Release notes and open source acknowledgment

The release notes list new features, improvements and known issues of the current software version, and describe the software installation.

The software uses several valuable open source software packages. An open source acknowledgment document provides verbatim license texts of the used open source software.

www.rohde-schwarz.com/firmware/areg800a

2.9 Application notes, application cards, white papers, etc.

These documents deal with special applications or background information on particular topics.

For some application sheets, see also:

www.rohde-schwarz.com/application/areg800a

2.10 Videos

Find various videos on Rohde & Schwarz products and test and measurement topics on YouTube: <https://www.youtube.com/@RohdeundSchwarz>



On the menu bar, search for your product to find related videos.



Figure 2-1: Product search on YouTube

3 Key features

The AREG is a new high performance Automotive Radar Echo Generator developed to meet demanding customer requirements. Offering excellent radar signal characteristics, wide radar signal bandwidth and straightforward and intuitive operation, the AREG makes radar signal generation fast and easy.

Outstanding key features of the AREG are:

- Simulation of echoes of stationary and highly dynamic objects
- Simulation of echoes with extremely short object distances
- Simulation of echoes of multiple independent objects
- Azimuth simulation with R&S QAT100
- IF signals with high instantaneous bandwidth
- Scalable solution from production test needs to road-ready vehicle simulators
- Fully harmonized with external frontends for a wide range of echo complexity
- Built-in real-time interface for dynamic scenario simulation
- Support of scenario simulation tools: DYNA4 from Vector, Open Simulation Interface (OSI)
- Synchronization of multiple instruments of the AREG and R&S QAT100

For more information, refer to the specifications document.

4 Preparing for use

Here, you can find basic information about setting up the product for the first time.

4.1 Lifting and carrying

For safety information, see "[Lifting and carrying the product](#)" on page 8.

Use the handles for lifting and carrying the AREG.

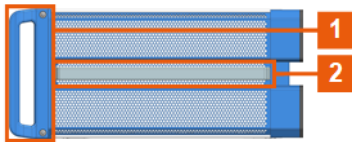


Figure 4-1: Instrument side view with front handles (1) and side handles (2)

4.2 Unpacking and checking

1. Unpack the AREG carefully.
2. Retain the original packing material. Use it to protect the control elements and connectors when transporting or shipping the AREG later.
See also chapter "Transporting" in the user manual.
3. Using the delivery notes, check the equipment for completeness.
4. Check the equipment for damage.

If the delivery is incomplete or equipment is damaged, contact Rohde & Schwarz.

4.3 Choosing the operating site

Specific operating conditions ensure proper operation and avoid damage to the product and connected devices. For information on environmental conditions such as ambient temperature and humidity, see the specifications document.

For safety information, see ["Choosing the operating site"](#) on page 8.

Electromagnetic compatibility classes

The electromagnetic compatibility (EMC) class indicates where you can operate the product. The EMC class of the product is given in the specifications document.

- Class B equipment is suitable for use in:
 - Residential environments
 - Environments that are directly connected to a low-voltage supply network that supplies residential buildings
- Class A equipment is intended for use in industrial environments. It can cause radio disturbances in residential environments due to possible conducted and radiated disturbances. It is therefore not suitable for class B environments. If class A equipment causes radio disturbances, take appropriate measures to eliminate them.

4.4 Setting up the AREG

See also:

- ["Setting up the product"](#) on page 9
- ["Intended use"](#) on page 7

4.4.1 Placing the AREG on a bench top

To place the product on a bench top

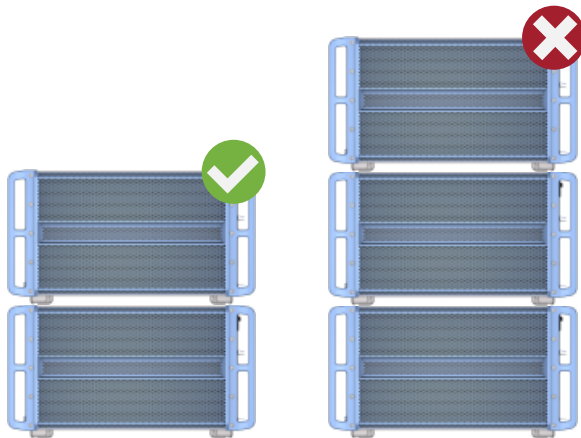
1. Place the product on a stable, flat and level surface. Ensure that the surface can support the weight of the product. For information on the weight, see the specifications document.

Setting up the AREG

2. **WARNING!** A stack of products can fall over and cause injury. Never stack more than two products. Otherwise, mount them in a rack.

Stack as follows:

- All products must have the same dimensions (width and length).
- Do not exceed a total load of 50 kg placed on the product at the bottom of the stack.



Left = Stacked correctly

Right = Stacked incorrectly, too many products

3. **NOTICE!** Overheating can damage the product.

Prevent overheating as follows:

- Keep a minimum distance of 10 cm between the fan openings of the product and any object in the vicinity to provide sufficient airflow and ventilation.
- Do not place the product next to heat-generating equipment such as radiators or other products.

4.4.2 Mounting the AREG in a rack

To prepare the rack

1. Observe the requirements and instructions in ["Setting up the product"](#) on page 9.
2. **NOTICE!** Insufficient airflow can cause overheating and damage the product. Design and implement an efficient ventilation concept for the rack.

To mount the AREG in a rack

1. Use an adapter kit that fits the dimensions of the AREG to prepare the instrument for rack mounting.
 - a) Order the rack adapter kit designed for the AREG.
For more information, refer to the specifications document.
 - b) Mount the adapter kit. Follow the assembly instructions provided with the adapter kit.
2. **WARNING!** The AREG is heavy. Lift the AREG with a lifting equipment, see ["Lifting and carrying the product"](#) on page 8.
Lift the AREG to shelf height.
3. Grab the handles at the front.
4. Push the AREG onto the shelf until the rack brackets fit closely to the rack.
5. Tighten all screws at the rack brackets with a tightening torque of 1.2 Nm to secure the AREG in the rack.

To unmount the AREG from a rack

1. Loosen the screws at the rack brackets.
2. **WARNING!** The AREG is heavy. Lift the AREG with a lifting equipment, see ["Lifting and carrying the product"](#) on page 8.
Bring the lifting equipment to shelf height.
3. Remove the AREG from the rack.
4. If placing the AREG on a bench top again, unmount the adapter kit from the AREG. Follow the instructions provided with the adapter kit.

4.5 Considerations for test setup

Cable selection and electromagnetic interference (EMI)

Electromagnetic interference (EMI) can affect the measurement results.

To suppress electromagnetic radiation during operation:

- Use high-quality shielded cables, for example, double-shielded RF, LAN and USB cables.

- Always terminate open cable ends.
- Ensure that connected external devices comply with EMC regulations.
- Use the cable G-SMA EF316 for the IF connection to the SMA connectors of the instrument and the R&S QAT100. The cable is included in the delivery of the R&S QAT100 and available under order number 1142.5889.00.
How to: ["To connect to SMA connectors"](#) on page 26
- Use the cables TRX CTRL Cable, IF TX CBL and IF RX CBL for connection of the instrument and the TRX frontend. The cables are included in the delivery of the TRX frontend and available under order number 1142.5889.00.
How to: ["To connect to SMA connectors"](#) on page 26

Signal input and output levels

Keep the signal levels within the specified ranges to avoid damage to the AREG and connected devices.

For more information, refer to the specifications document.

Preventing electrostatic discharge (ESD)

Electrostatic discharge is most likely to occur when you connect or disconnect a DUT.

- **NOTICE!** Electrostatic discharge can damage the electronic components of the product and the device under test (DUT).

Ground yourself to prevent electrostatic discharge damage:

- a) Use a wrist strap and cord to connect yourself to ground.
- b) Use a conductive floor mat and heel strap combination.

Over-the-air (OTA) tests

For over-the-air (OTA) tests, operate the instrument in a shielded environment.

For more information, refer to the specifications document.

4.6 Connecting to power

For safety information, see ["Connecting to power"](#) on page 9.

Connecting to LAN

1. Plug the AC power cable into the AC power connector on the rear panel. Only use the AC power cable delivered with the AREG.
2. Plug the AC power cable into a power outlet with ground contact.
The required ratings are listed next to the AC power connector.

For more information, refer to the specifications document.

4.7 Connecting to LAN

Network environment

Before connecting the product to a local area network (LAN), consider the following:

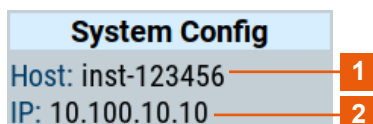
- Install the latest firmware to reduce security risks.
- For internet or remote access, use secured connections if applicable.
- Ensure that the network settings comply with the security policies of your company. Contact your local system administrator or IT department before connecting your product to your company LAN.
- When connected to the LAN, the product may potentially be accessed from the internet, which may be a security risk. For example, attackers might misuse or damage the product.

To connect to LAN

- Connect the LAN socket on the rear panel via an RJ-45 cable to the LAN.

Using DHCP (dynamic host configuration protocol), the AREG assigns the IP address automatically.

If connected to the LAN, the AREG displays the IP address and the hostname at the left bottom of the home screen.



1 = Hostname
2 = IP address

If disconnected from the LAN, the AREG displays the IP address *0.0.0.0*.

See also chapter "Setting up remote control in the user manual.

4.8 Connecting USB devices

You can connect or disconnect all USB devices from the AREG during operation.

To connect USB storage devices

USB storage devices, such as memory sticks, allow data transfer from or to the AREG. You can also use them for firmware updates.

1. Connect the USB storage device to any of the USB connectors.
2. Connect the device directly, without a connecting cable.
Connecting cables can cause electromagnetic radiation and impair a measurement result.

To connect USB devices with an external power supply

1. **NOTICE!** Connected devices with external power supply can feed back current into the 5 V power supply of the USB interface and thus damage the AREG.

Ensure that there is no connection between the positive pole of the power supply and the +5 V power pin of the USB interface (VBUS).

2. Connect the USB storage device to any of the USB connectors on the front panel or rear panel.

To connect a keyboard

- Connect the keyboard to any of the USB connectors on the front panel or rear panel.

When connected, the AREG detects the keyboard automatically. A detected keyboard has the default layout English – US.

To connect a mouse

- Connect the mouse to any of the USB connectors.

Connecting to RF coaxial connectors

When connected, the AREG detects the mouse automatically.

To connect power sensors

Connect power sensors of the R&S NRP families to any of the USB connectors on the front panel or rear panel.

See chapter "Using power sensors" in the user manual.

4.9 Connecting to RF coaxial connectors

Here, you find information on how to prepare and to connect to RF coaxial connectors of the AREG. Use these RF connectors, for example, for output of the RF signal or for input of an external reference signal.

To prepare for connecting

1. **NOTICE!** Damaged or not clean connections can lead to RF insertion loss and mismatch, and even premature wear of the connectors.
Before connecting to the port, inspect the RF connector visually. Check that it is clean, undamaged and mechanically compatible.
2. **NOTICE!** DC voltage at the RF connector can damage the instrument. Never apply DC voltage to the RF input connectors.
Make sure that the values are within the DC limits given in the specifications document.
3. If your test setup has a DC component at the RF input, insert a DC blocker.
4. Use a high-quality RF cable that matches the RF connector type. See also ["Cable selection and electromagnetic interference \(EMI\)"](#) on page 20.
5. You can connect to two kinds of connectors:
 - ["To connect to screwable connectors"](#) on page 24
 - ["To connect to pluggable connectors"](#) on page 25

To connect to screwable connectors

- **NOTICE!** Excessive tightening can damage the connectors.

To connect the cable with the connector, proceed as follows:

- a) Carefully align the connector of the cable and the connector along a common axis.

Connecting to RF coaxial connectors

- b) Mate the connectors along the common axis until the male pin of the inner connector engages with the female socket of the outer connector.
- c) Turn the nut of the outer connector until the connectors are firmly coupled.
- d) Using a calibrated torque wrench torque the nut to the limit as in the table below. Hold the opposite connector part stationary with a spanner.

The AREG provides screwable RF connectors as in [Table 4-1](#).

Table 4-1: Connector name, type, size, torque limit and nut opening

Connector		Torque limit		Nut opening	
Type	Name	lb-Inch	Nm	Inch	mm
SMA	Ext Clk In	5	0.56	5/16	8
	Sync In				
	Sync Out				
	Aux IF In				
	Aux IF Out				
	Rx IF In				
	Rx IF Out				
	Tx IF In				
	Tx IF Out				

To connect to pluggable connectors

The AREG provides pluggable Bayonet Neill-Concelman (BNC) connectors.

- To connect the RF cable with the BNC connector, proceed as follows:
 - a) Carefully align the connector of the cable and the BNC connector along a common axis.
 - b) Mate the connectors along the common axis until the male pin of the connector of the cable engages with the female socket of the BNC connector.

For more information on handling and maintaining coaxial RF connectors, see the application note [1MA99](#).

For information on mounting test port adapters onto the RF connector, see the application note [1MA100](#).

4.10 Connecting the R&S QAT100

The AREG can generate radar signals when connected to an external frontend, e.g. the R&S QAT100. This section provides information on relevant connections between AREG and the R&S QAT100.

To connect to SMA connectors

For IF signal input and output the AREG provides SMA connectors.

1. Before connecting, prepare the SMA connectors on the rear panel. See ["To prepare for connecting"](#) on page 24.
2. Connect to the SMA connectors of the AREG with the R&S QAT100. See ["To connect to screwable connectors"](#) on page 24.

You have several options. [Table 4-2](#) provides an example on relevant connections.

To connect to System Control

- Connect the socket of the "System Control" connector with an RJ-45 cable to the "LAN" connector of the R&S QAT100. See [Chapter 4.7, "Connecting to LAN"](#), on page 22.

[Table 4-2](#) provides an overview on relevant connections.

Table 4-2: Connection between AREG and R&S QAT100

Signal (direction)	AREG connector	R&S QAT100 connector
IF out (Tx)	"Tx IF Out 1"	"Tx Σ "
IF in (Rx)	"Rx IF In 1"	"Rx Select"
Control (LAN)	"System Control"	"LAN"

To disconnect the R&S QAT100

1. **NOTICE!** If you connect or disconnect the control cable while the base unit is powered on, you can damage the AREG.
Switch off the AREG, see ["To shut down the product"](#) on page 32.
2. Disconnect the SMA connections between AREG and R&S QAT100:

Connecting to Digital IO DD

- a) Untorque the nut using a calibrated torque wrench. Hold the opposite connector part stationary with a spanner.
 - b) Turn the nut of the outer connector until the connectors are detached.
3. To disconnect the connection cable:
 - a) Hold the connector at the sleeve.
 - b) To release the lock, pull the sleeve without turning.
Do not pull the cable.

4.11 Connecting to Ref In/Ref Out

For connection, the AREG provides BNC connectors on the rear panel. See ["To connect to pluggable connectors"](#) on page 25.

4.12 Connecting to Digital IO DD

To connect to the QSFP+ interface

1. For connection, use the cable R&S DIGIQ-HS.
See ["Cable selection and electromagnetic interference \(EMI\)"](#) on page 20.
2. Hold the QSFP+ plug of the cable by its panes.
3. Turn the QSFP+ cable, so that the release tab shows upwards.
4. Insert and push the QSFP+ plug into the QSFP+ cage.

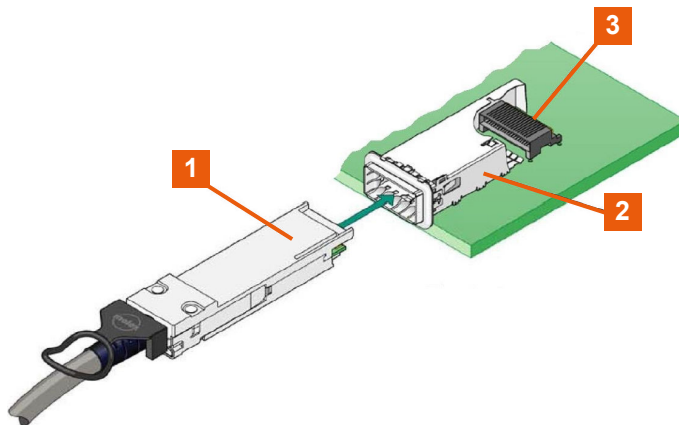
If you hear a clicking sound, the plug engaged correctly with the cage and the cable is connected to the interface.

To disconnect from the QSFP+ interface

1. **NOTICE!** If you pull the cable, you can damage the cable and the Digital IO DD connector.
Pull the release tab.
2. Pull the QSFP+ plug out of the QSFP+ cage.

About the QSFP+ interface

The "Digital IO DD" connector is a QSFP+ (quad small form factor pluggable) interface or QSFP28 interface. The connector socket has two components: a QSFP+ cage and a QSFP+ connector. The QSFP+ cable is equipped with the QSFP+ plug. The interface supports a maximum bandwidth of up to 50 Gsample/s with active optical cables.



- 1 = QSFP+ plug
- 2 = QSFP+ cage
- 3 = QSFP+ connector

4.13 Connecting to Aux IF In/Aux IF Out

See ["To connect to SMA connectors"](#) on page 26.

4.14 Connecting a TRX frontend

The TRX frontend is delivered with two IF cables (IF TX CBL/IF RX CBL) and one control cable (TRX CTRL Cable). These cables are calibrated and dedicated to the particular TRX frontend. Each cable has its own serial number, printed on it.

To connect the "IF TX CBL"/"IF RX CBL" cables

1. Take the **"IF TX CBL"** cable delivered with the TRX frontend.
See ["To check the serial number of the cables"](#) on page 30.

Connecting a TRX frontend

2. Connect the "Tx IF Out" connector of the base unit to "Tx IF In" connector of the TRX frontend.
Follow the instructions in [Chapter 4.13, "Connecting to Aux IF In/Aux IF Out"](#), on page 28.
This connection carries the IF signal for the transmitting antenna and the reference signal for the TRX frontend.
3. Take the **"IF RX CBL"** cable delivered with the TRX frontend.
4. Connect the "Rx IF Out" connector of the TRX frontend to "Rx IF In" connector of the base unit.
Follow the instructions in [Chapter 4.13, "Connecting to Aux IF In/Aux IF Out"](#), on page 28.
This connection carries the IF signal of the receiving antenna for the base unit.
5. **NOTICE!** Excessive bending can damage the cables.
Bend the "IF TX CBL"/"IF RX CBL" to a max. bending radius of 50 mm.

To connect the TRX CTRL Cable cable

1. **NOTICE!** If you connect or disconnect the control cable while the base unit is powered on, you can damage the AREG.
Switch off the base unit.
See ["To shut down the product"](#) on page 32.
2. Take the **TRX CTRL Cable** cable delivered with the TRX frontend.
See ["To check the serial number of the cables"](#) on page 30.
3. Hold the connector at the sleeve so that the red dot is upwards.
4. Align the connector of the cable along a common axis.
5. Plug the control cable at the "Control" connector at the base unit.
The sleeve locks.
6. Plug the control cable at the "Control" connector at the TRX frontend.
The sleeve locks.
7. **NOTICE!** Excessive bending can damage the cables.
Bend the "TRX CTRL Cable" to a maximum bending radius of 50 mm.

To operate the TRX frontend

1. Switch on the base unit.
See ["To shut down the product"](#) on page 32.
2. **CAUTION!** Hot surface, see ["Meaning of safety labels"](#) on page 10.
Hold the TRX frontend by its sides.
Do not touch the waveguide surface. Depending on the ambient temperature, this surface can become hot.
3. **NOTICE!** Overheating can damage the product.
Prevent overheating as follows:
 - Do not cover the TRX frontend module with thermally insulating material while the AREG is switched on.
 - See also [Chapter 4.4.1, "Placing the AREG on a bench top"](#), on page 18.

To disconnect the TRX frontend

1. **NOTICE!** If you connect or disconnect the control cable while the base unit is powered on, you can damage the AREG.
Switch off the base unit.
See ["To shut down the product"](#) on page 32.
2. To disconnect the "IF TX CBL"/"IF RX CBL" cables, proceed as follows for each of the four connections:
 - a) Untorque the nut using a calibrated torque wrench. Hold the opposite connector part stationary with a spanner.
 - b) Turn the nut of the outer connector until the connectors are detached.
3. To disconnect the "TRX CRTL Cable":
 - a) Hold the connector at the sleeve.
 - b) To release the lock, pull the sleeve without turning.
Do not pull the cable.

To check the serial number of the cables

1. Switch on the base unit.
See [Chapter 4.15, "Switching on or off"](#), on page 31.
2. On the front panel, press [Setup].
3. Select "Setup" > "Instrument Assembly" > "Hardware Config" > "RF Assembly".

4. For each cable, compare the displayed serial number with the serial number printed on the cable.

4.15 Switching on or off

The following table provides an overview of power states, LEDs and positions of the power switch.

Table 4-3: Overview of power states

State	LED	Position of power switch
Off	 gray	[0]
Standby	 orange	[I]
Ready	 green	[I]

To switch on the AREG

The AREG is off but connected to power. See [Chapter 4.6, "Connecting to power"](#), on page 21.

1. Set the switch on the power supply to position [I] on the rear panel.
The LED of the [On/Standby] key on the front panel is orange.

2. Press the [On/Standby] key on the front panel.

The LED changes to green. The AREG boots.

After booting, the AREG starts up displaying the home screen on the screen on the front panel.

To check startup functions

When starting for the first time, the AREG starts with the default settings. When restarting, the AREG recalls the instrument configuration before shut-down.

See chapter "Saving and recalling instrument settings" in the user manual.

When switched on, the AREG monitors the main functions and logs erroneous functions. See the following:

See chapter "Querying notifications" in the user manual.

To shut down the product

The product is in the ready state.

- ▶ Press the [On/Standby] key.

The operating system shuts down. The LED changes to orange.

In the standby state, the power switch circuits are active. To deactivate them, disconnect the instrument from the power supply.

To disconnect from power

The AREG is in the standby state.

1. **NOTICE!** Risk of data loss. If you disconnect the product from power when it is in the ready state, you can lose settings and data. Shut it down first.
Set the toggle switch on the power supply to position [0].
The LED of the [On/Standby] key is switched off.
2. Disconnect the AREG from the power source.

5 Instrument tour

This chapter explains the control elements and the connectors of the AREG. The views of the front panel and the rear panel help you to get familiar with the instrument and to perform first steps. For specifications of the interfaces, see the specifications document.

The meanings of the labels on the AREG are described in [Chapter 1.2, "Labels on AREG"](#), on page 11.

5.1 Front panel tour

This section provides an overview of the control elements and connectors on the front panel of the AREG. On the [rear panel](#), you find all further connectors of the unit. The user interface can be displayed on a remote PC station used to manually remote control the instrument.

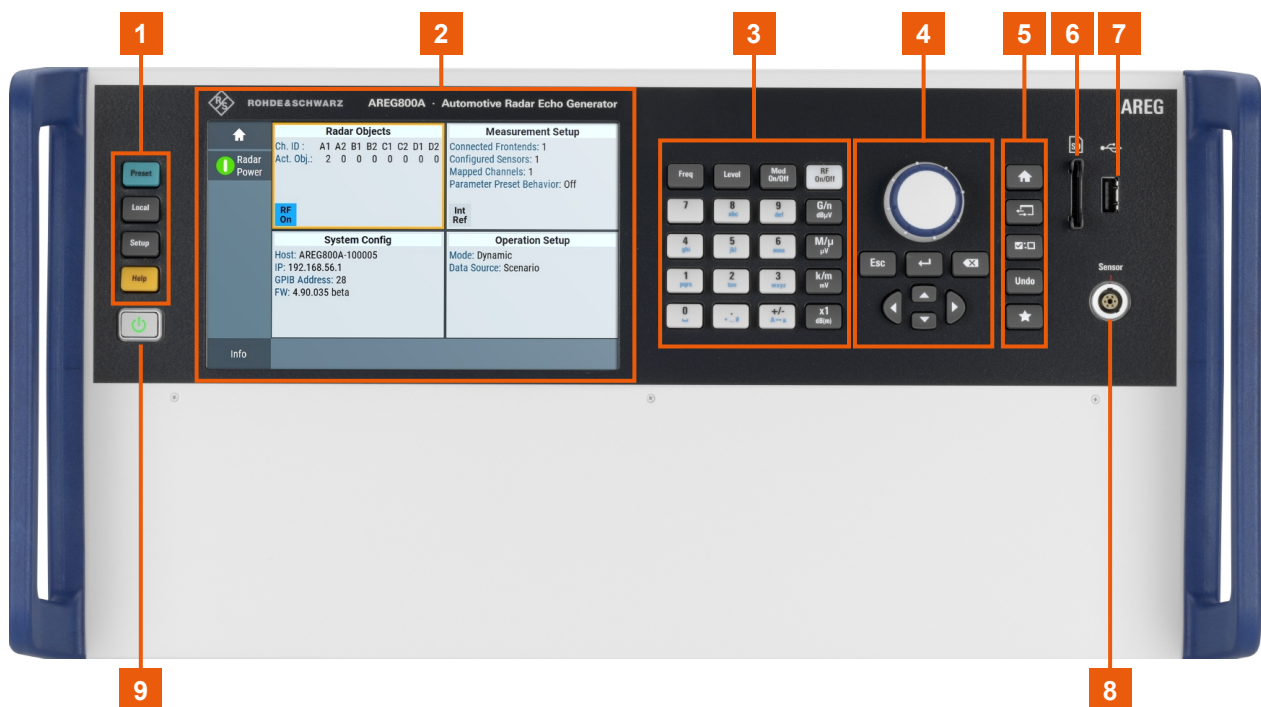


Figure 5-1: AREG front panel controls and connectors

- 1 = [Utility keys](#), page 35
- 2 = [Touchscreen](#), page 34
- 3 = [Function keys](#), page 36 and [Keypad](#), page 36

- 4 = [Navigation controls, page 37](#)
- 5 = [Display keys, page 38](#)
- 6 = [SD card slot, page 39](#)
- 7 = [USB, page 39](#)
- 8 = [Sensor, page 39](#)
- 9 = [On/Standby, page 35](#)

5.1.1 Touchscreen

The most important settings are displayed on the screen on the front panel. The screen display also provides status and setting information and allows you to quickly reconfigure the signal flow. The screen is touch-sensitive, offering an alternative means of user interaction for quick and easy handling of the instrument.

The screen at the front panel is the graphical user interface. It shows the settings dialogs and parameters, see [Chapter 7.2, "Understanding the display information"](#), on page 52. The screen is touch-sensitive, offering an alternative means of user interaction for quick and easy handling of the instrument.

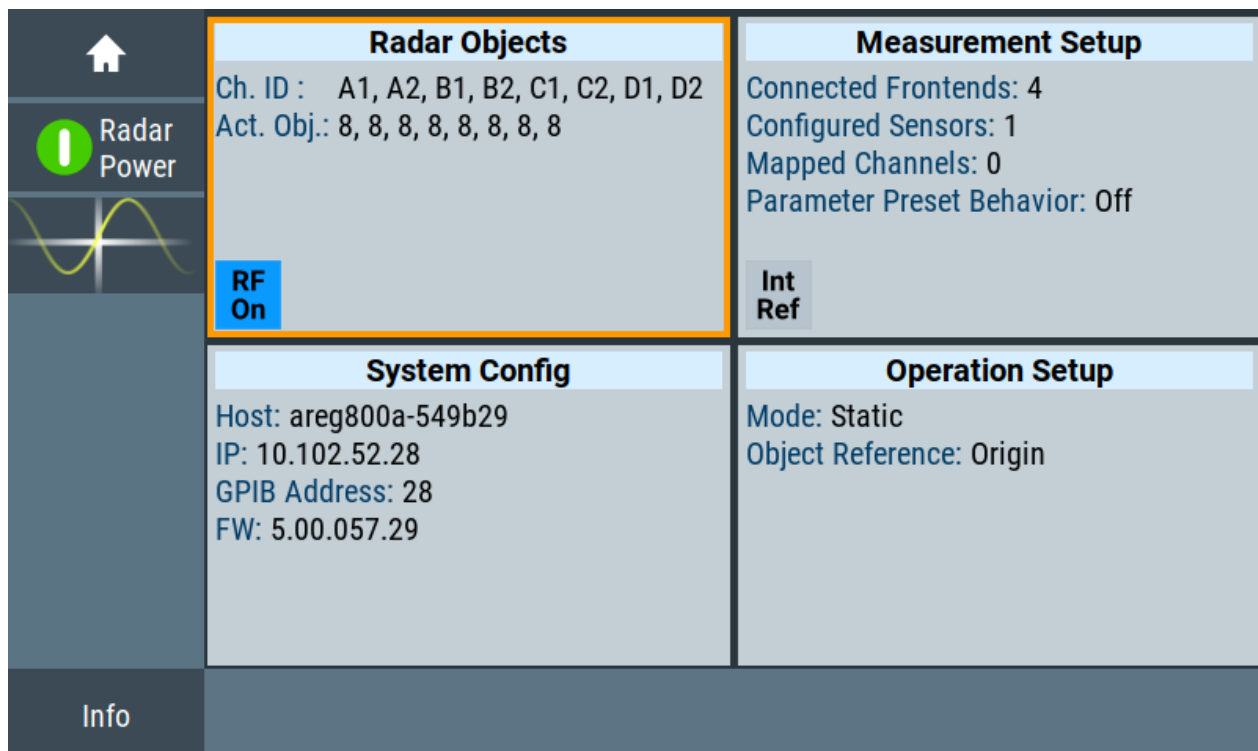


Figure 5-2: Touchscreen

Any user interface elements that react to a click by a mouse pointer also react to a tap on the screen, and vice versa. Using the touchscreen, you can perform the following tasks (among others) by the tap of your finger:

- Changing a setting
- Selecting new settings
- Scrolling through the list of parameters
- Saving or recalling settings
- Routing the signal flow
- Opening and closing dialogs

See also:

- [Chapter 7, "Operating the instrument"](#), on page 51, for operating the touchscreen.
- Chapter "Cleaning" in the user manual, for cleaning the touchscreen.

5.1.2 Keys

5.1.2.1 Utility keys

The utility keys cause the AREG to return to a defined instrument state and provide information on the instrument and assistance.

Table 5-1: Utility keys

Utility key	Assigned functions
[Preset]	Sets the instrument to a defined state
[Local]	Switches from remote control to local (manual) control
[Setup]	Accesses the general instrument settings
[Help]	Displays context-sensitive help text

5.1.2.2 On/Standby

The [On/Standby] key switches the instrument from the standby to the ready state or vice versa.

The LED above the [On/Standby] key indicates the instrument state, see [Chapter 4.15, "Switching on or off"](#), on page 31.

5.1.2.3 Function keys

Function keys provide access to the most common generator settings and functions. You can find a detailed description of the corresponding functions in the user manual.

Table 5-2: Function keys

Function key	Assigned functions
[Freq]	Requires a connected external frontend, e.g., TRX frontends or an R&S QAT100. Opens the "Frontend Config" dialog and selects the "Frontend Center Frequency" of the connected external frontend. If several external frontends are connected, switches between the "Frontend Center Frequency" setting of the connected external frontends.
[Level]	Opens the "Radar Power" dialog, switches on or switches off selected radar channels.
[Mod On/Off]	Opens and closes the "Objects" dialog. In particular, selects the "Channel x" tab of the first available radar channel.
[RF On/Off]	Activates and deactivates all available radar objects of all radar channels.

5.1.2.4 Keypad

The keypad is used to enter alphanumeric parameters, including the corresponding units. It contains the following keys:

Table 5-3: Keys on the keypad

Type of key	Description
Alphanumeric keys	Enter numbers and (special) characters in edit dialog boxes.
Decimal point	Inserts a decimal point "." at the cursor position.
Sign key	Changes the sign of a numeric parameter. For an alphanumeric parameter, inserts a "-" at the cursor position.
Unit keys (G/n dBμV, M/μ μV, k/m mV and x1 dB(m))	These keys add the selected unit to the entered numeric value and complete the entry. For level entries (e.g. in dB) or dimensionless values, all units have the value "1" as multiplying factor. Thus, they have the same function as an [Enter] key.

5.1.2.5 Navigation controls

The navigation controls include a rotary knob, navigation keys, and the display keys. They allow you to navigate within the display or within dialog boxes.

Rotary knob

The rotary knob has several functions:

- Increments (clockwise direction) or decrements (counterclockwise direction) the instrument parameter at a defined step width for a numeric entry.
- Moves the selection, e.g. to a function block in the block diagram
- Shifts the selection bar within focused areas (e.g. lists).
- Acts like the [Enter] key, when it is pressed.

Editing keys

Editing keys enable you to confirm an entry, delete individual characters, or exit the current operation.

Table 5-4: Editing keys

Type of key	Description
[Esc] key	Closes all kinds of dialog boxes, if the edit mode is not active. Quits the edit mode, if the edit mode is active. In dialog boxes that contain a "Cancel" button it activates that button. For "Edit" dialog boxes, the following mechanism is used: • If data entry has been started, it retains the original value and closes the dialog box. • If data entry has not been started or has been completed, it closes the dialog box.
[Enter] key	Has the same effect as pressing the rotary knob • Concludes the entry of dimensionless entries. The new value is accepted. • With other entries, this key can be used instead of the default unit key. • In a dialog box, selects the default or focused element. • Calls the next dialog level. • Confirms and closes open input windows.
[Backspace] key	Deletes the character to the left of the cursor in editing mode.

Navigation keys

The navigation keys can be used alternatively to the rotary knob to navigate through dialog boxes, diagrams, or tables.

Table 5-5: Navigation keys

Type of key	Description
[Up/Down] key	The [Up] and the [Down] key does the following: • In a numeric edit dialog box, increase or decrease the instrument parameter. • In a list, scroll forward and backward through the list entries. • In a table, move the selection bar vertically. • In windows or dialog boxes with vertical scrollbar, move the scrollbar.
[Left/Right] key	The [Left] and the [Right] key does the following: • In an alphanumeric edit dialog box, move the cursor. • In a list, scroll forward and backward through the list entries. • In a table, move the selection bar horizontally. • In windows or dialog boxes with horizontal scrollbar, move the scrollbar.

5.1.2.6 Display keys

The display keys arrange different windows on the display.

Table 5-6: Display keys

Display key	Assigned functions
[Home]	Returns to the initial feature screen.
[Next window]	Toggles between the entry fields in the taskbar.
[On/Off]	• Switches highlighted elements or a function block on and off. • Switches between two or more settings, e.g. items of selection lists. At the end of a list, the cursor is set on the first entry again.
[Undo]	Reverts the last operation.
[User]	Adds a parameter to the user menu for quick access.

5.1.3 Connectors

Various interface connectors are on the front panel of the AREG.

SD card slot

SD card slot.

USB

Female USB (universal serial bus) 2.0 connector of type A, to connect devices like external USB devices, a mouse or a keyboard for enhanced operation.

You can also connect R&S NRP power sensors (with an R&S NRP-Z4 or an R&S NRP-ZKU adapter cable) for external power measurements. Also, you can connect a memory stick for software updates and data exchange.

Further USB connectors of type A are available on the rear panel.

How to: [Chapter 4.8, "Connecting USB devices"](#), on page 23

Sensor

Connector for R&S NRP sensors.

The connector is a six-pole ODU Mini-Snap® series B connector.

A power sensor is connected to the AREG by inserting the male connector. To disconnect, pull the connector by its sleeve. You cannot disconnect the sensor simply by pulling at the cable or the rear part of the connector.

The AREG supports the use of R&S NRP power sensors in various ways including the use as a power viewer.

How to: Chapter "Using Power Sensors" in the user manual

5.2 Rear panel tour

This section provides an overview of the connectors on the rear panel of the instrument. For technical data of the connectors, refer to the specifications document.



Figure 5-3: AREG rear panel

- 1 = [Connectors for control, synchronization and multi-purpose signals, page 41](#)
- 2 = [Power switch, power supply, monitor, LAN and USB connectors, page 43](#)
- 3 = [Digital IO DD connectors and instrument boards, page 44](#)

5.2.1 Control, synchronization and multi-purpose connectors

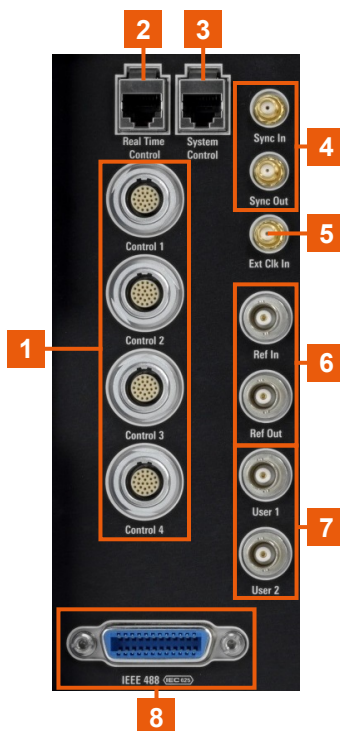


Figure 5-4: Connectors for control, synchronization and multi-purpose signals

- 1 = [Control](#), page 41
- 2 = [Real Time Control](#), page 41
- 3 = [System Control](#), page 42
- 4 = [Sync In, Sync Out](#), page 42
- 5 = [Ext Clk In](#), page 42
- 6 = [Ref In/Ref Out](#), page 42
- 7 = [User x](#), page 42
- 8 = [IEEE 488](#), page 42

Control

Four propriety bus connectors "Control 1" to "Control 4" for control of TRX frontends.

How to: [Chapter 4.14, "Connecting a TRX frontend"](#), on page 28

Real Time Control

RJ-45 connector to transfer real-time data in dynamic radar scenarios. Real-time data transfer is required for hardware in the loop (HiL) or vehicle in the loop (ViL) testing.

System Control

RJ-45 connector to control R&S QAT100 frontends.

How to: [Chapter 4.7, "Connecting to LAN"](#), on page 22

Sync In, Sync Out

SMA female connectors for synchronization of multi-device setups including more instruments.

Note: When using a single AREG in your test setup, make sure that the Sync In and Sync Out connectors are connected with the bracket provided with the delivery.

How to: ["To connect to SMA connectors"](#) on page 26

Ext Clk In

SMA female connector for input of an external reference signal.

How to: ["To connect to SMA connectors"](#) on page 26

Ref In/Ref Out

Input/output for external reference signal.

BNC connectors for 10 MHz reference signals.

How to: [Chapter 4.11, "Connecting to Ref In/Ref Out"](#), on page 27

User x

BNC multipurpose connectors for defining output signals.

[Table 5-7](#) lists the signals assigned to the User x connectors in the default instrument state.

Table 5-7: Default configuration of the User x connectors

Connector	Direction	Default assigned signal
"User 1"/"User 2"	"Output"	"Object Marker"
	"Not used"	"None"

See also chapter "Global connectors settings" in the user manual.

IEEE 488

General purpose interface bus (GPIB) interface for remote control of the instrument. The interface complies with the standards IEC 625, IEEE 488 and SCPI.

Use this interface to connect a computer for remote control of the AREG. See also the following sections:

- Chapter "Network operation and remote control" in the user manual
- "Annex: GPIB interface" in the user manual

5.2.2 Monitor, LAN, USB and power connectors

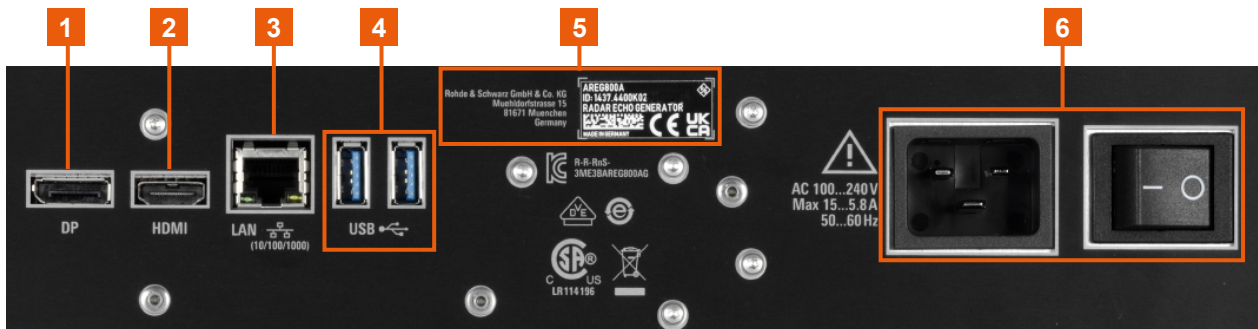


Figure 5-5: Power switch, power supply, monitor, LAN and USB connectors

- 1, 2 = DP, HDMI, [page 43](#)
- 3 = LAN, [page 43](#)
- 4 = USB, [page 43](#)
- 5 = Serial number
- 6 = AC power supply connector and switch, [page 43](#)

DP, HDMI

Display port (DP) and HDMI port for connecting external monitor.

LAN

RJ-45 connector to connect the AREG to a LAN for remote control, remote operation, and data transfer.

How to: [Chapter 4.7, "Connecting to LAN"](#), on page 22

USB

Two female USB (universal serial bus) 3.0 connectors of type A. You can connect devices like a keyboard, a mouse, a memory stick or the R&S NRP-Z4 cable for the R&S NRP power sensors.

How to: [Chapter 4.8, "Connecting USB devices"](#), on page 23

AC power supply connector and switch

Mains power switch for performing the following tasks:

- Connecting the internal power supply to the power source

- Disconnecting the internal power supply from the power source

How to: [Chapter 4.6, "Connecting to power"](#), on page 21.

5.2.3 DIG I/Q and RF connectors



Figure 5-6: Digital IO DD connectors and instrument boards

- 1 = Digital IO DD, page 44
- 2, 10 = Aux IF In, page 44
- 3, 9 = Tx IF Out, page 45
- 4, 8 = Aux IF Out, page 45
- 5, 7 = Rx IF In, page 45
- 6 = Four instrument boards "A" to "D" (4x R&S AREG8-B9)

Digital IO DD

Digital I/Q connectors "Digital IO DD 1" to "Digital IO DD 4" provided for future use; one connector per instrument board (R&S AREG8-B9).

Aux IF In

SMA female connectors "Aux IF In 1" and "Aux IF In 2" for input of an additional interferer to the generated echo signal; two connectors per instrument board (R&S AREG8-B9).

How to: ["To connect to SMA connectors"](#) on page 26

Aux IF Out

SMA female connectors "Aux IF Out 1" and "Aux IF Out 2" for output of IF signals for signal analysis; two connectors per instrument board (R&S AREG8-B9).

How to: ["To connect to SMA connectors"](#) on page 26

Rx IF In

SMA female connectors "Rx IF In 1" and "Rx IF In 2" for input of the intermediate frequency (IF) signal from the external frontend to the AREG (Rx); two connectors per instrument board (R&S AREG8-B9).

How to: ["To connect to SMA connectors"](#) on page 26

Tx IF Out

SMA female connectors "Tx IF Out 1" and "Tx IF Out 2" for output of the intermediate frequency (IF) signal from AREG to the external frontend (Tx); two connectors per instrument board (R&S AREG8-B9).

How to: ["To connect to SMA connectors"](#) on page 26

6 Trying out the instrument

This chapter introduces the first steps with the AREG. It shows how to operate and configure the instrument using simple examples. The complete description of the functionality and its usage is given in the AREG user manual.

Prerequisites

- The AREG is set up, connected to power, and started up.
See [Chapter 4, "Preparing for use"](#), on page 17.
- The AREG is connected to an external frontend, e.g. the R&S QAT100.
See [Chapter 4.10, "Connecting the R&S QAT100"](#), on page 26.
- The AREG is manually operated via the touchscreen.

Try out the following:

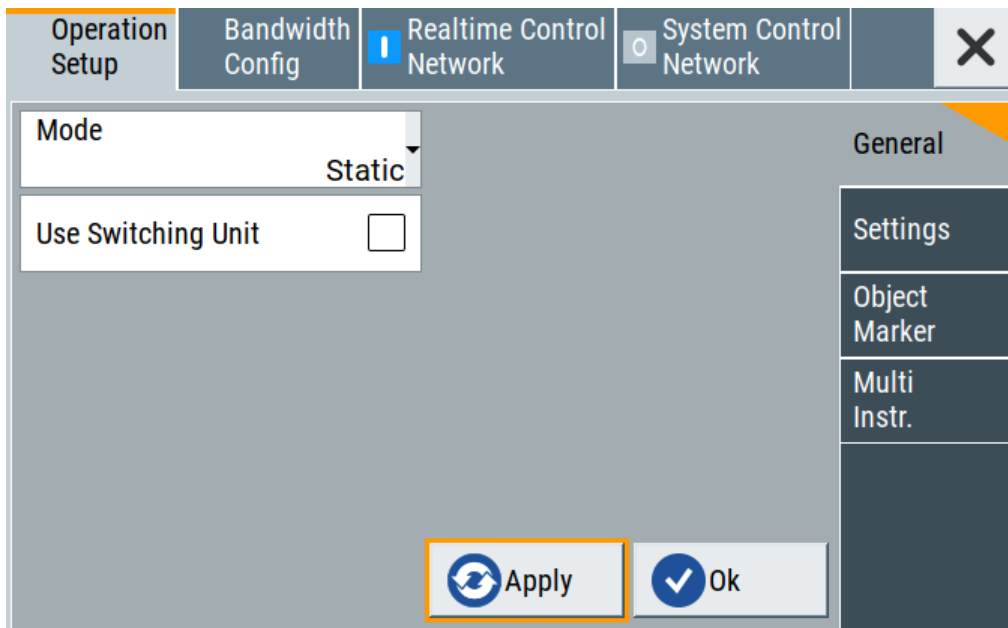
- [Configuring static operation setups](#).....46
- [Saving and recalling settings](#)..... 48

6.1 Configuring static operation setups

To specify characteristics of the radar object

1. In the tile diagram, select "Operation Setup".
2. Specify the radar object movement of the requirement:
 - Select "Mode > Static" for simulation of static radar objects.
 - Select "Mode > Dynamic" for simulation of dynamic radar objects.
3. Specify the reference of the radar object:
 - Select "Object Reference > Origin", if the AREG is the object reference.
 - Select "Object Reference > Mapped Sensor", if a sensor is the object reference. This sensor is connected

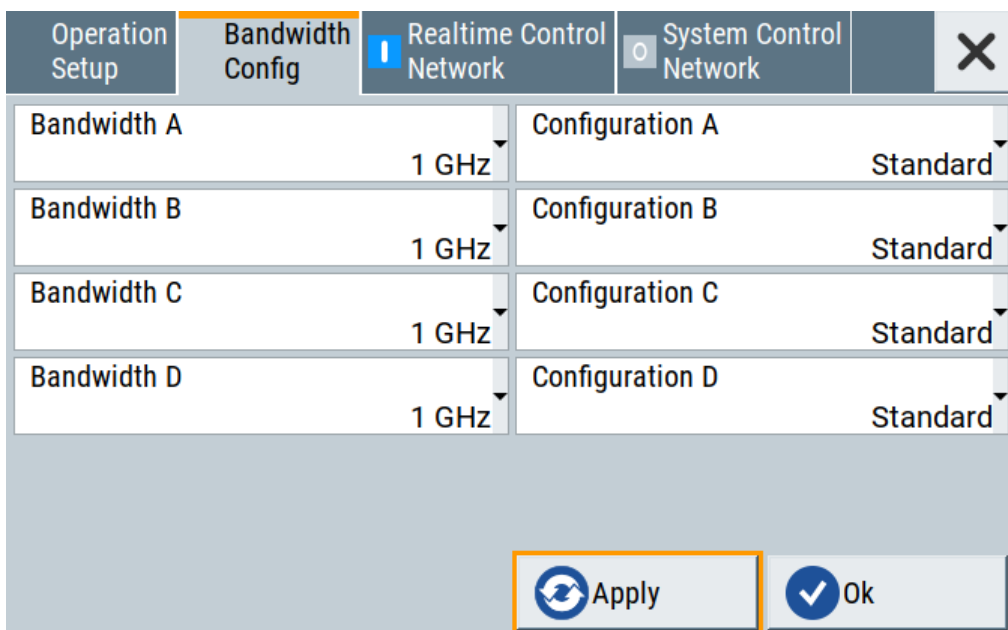
- Click "Apply".



The settings are applied.

To specify the radar channel bandwidth

- In the tile diagram, select "Operation Setup > Bandwidth Config".
- Set the bandwidths of the radar channels, e.g. "Bandwidth A > 1 GHz".
- Click "Apply" and "Ok".



The bandwidth settings for an external frontend connected to board "A" are applied.

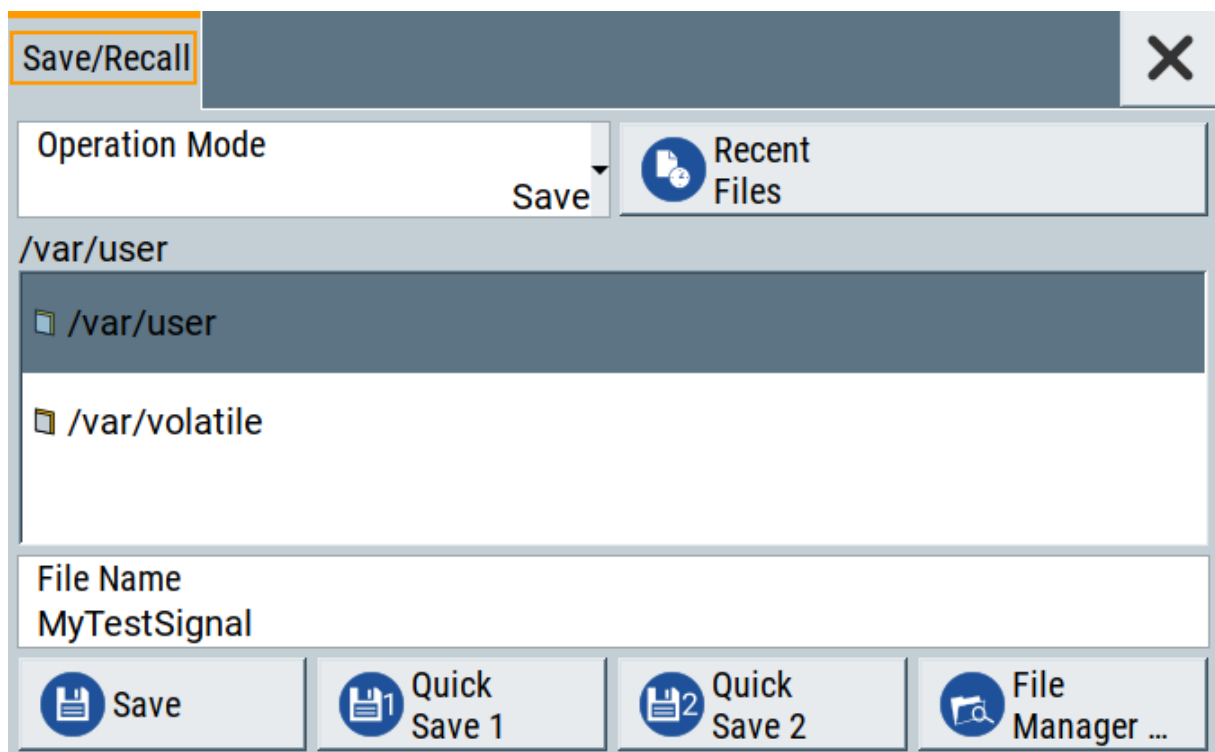
6.2 Saving and recalling settings

To restore the results of our measurements later, we save the instrument settings in a file.

To save the instrument settings in a file

We assume, a test configuration as described in [Chapter 6.1, "Configuring static operation setups"](#), on page 46.

1. Press the [Setup] key on the front panel.
2. In the "Setup" menu, select "Settings > Save/Recall".
3. In the "Save/Recall" dialog, select "Operation Mode > Save".



4. Select the "Filename" input field, use the on-screen keyboard, and enter *MyTestSignal*.

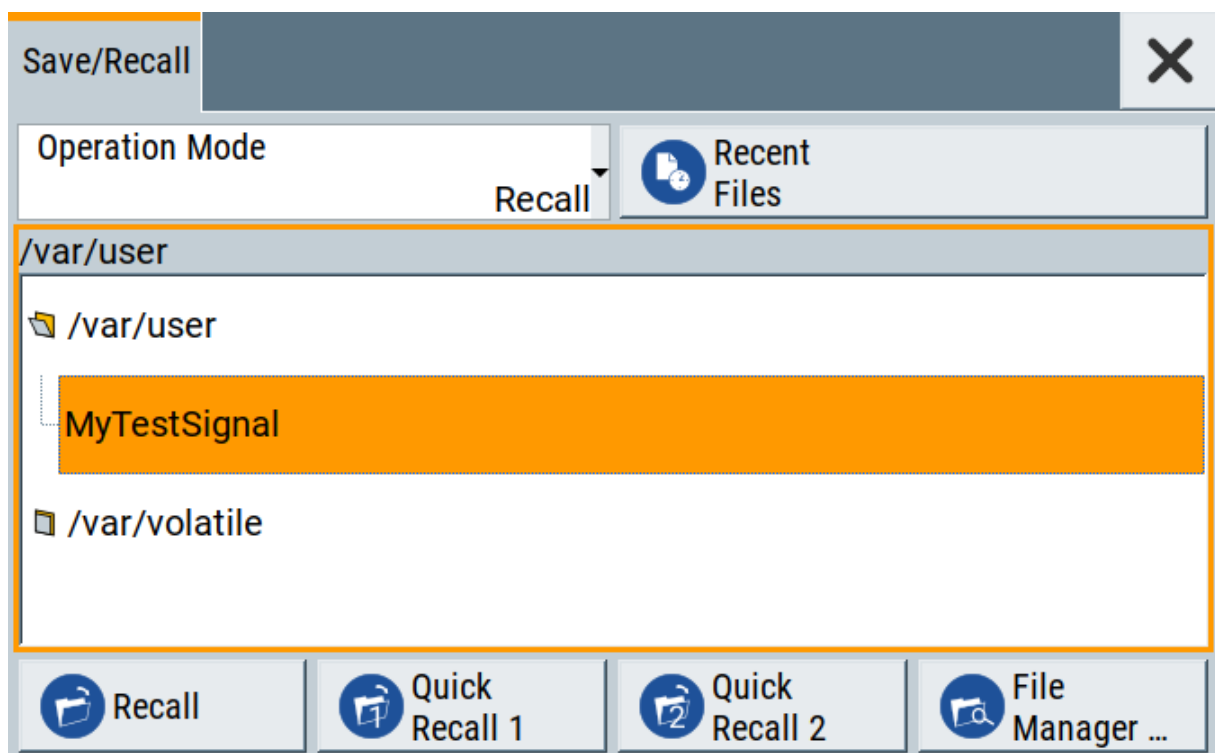
5. Click "Save".

The file `MyTestSignal.savrc1.txt` is stored in the default directory `/var/user/`.

To load saved instrument settings

You can restore the settings to the instrument at any time using the settings file.

1. Press the [Preset] button to restore the default instrument settings so you can check that the stored user settings are restored afterwards.
2. Press the [Setup] key on the front panel.
3. In the "Setup" menu, select "Settings > Save/Recall".
4. In the "Save/Recall" dialog, select "Operation Mode > Recall".
Navigate to the directory the file is saved in and select the `MyTestSignal` file.



5. Click "Recall".

All instrument settings are restored and the display resembles [Chapter 6.1, "Configuring static operation setups"](#), on page 46, which shows the instrument display right before the settings were saved.

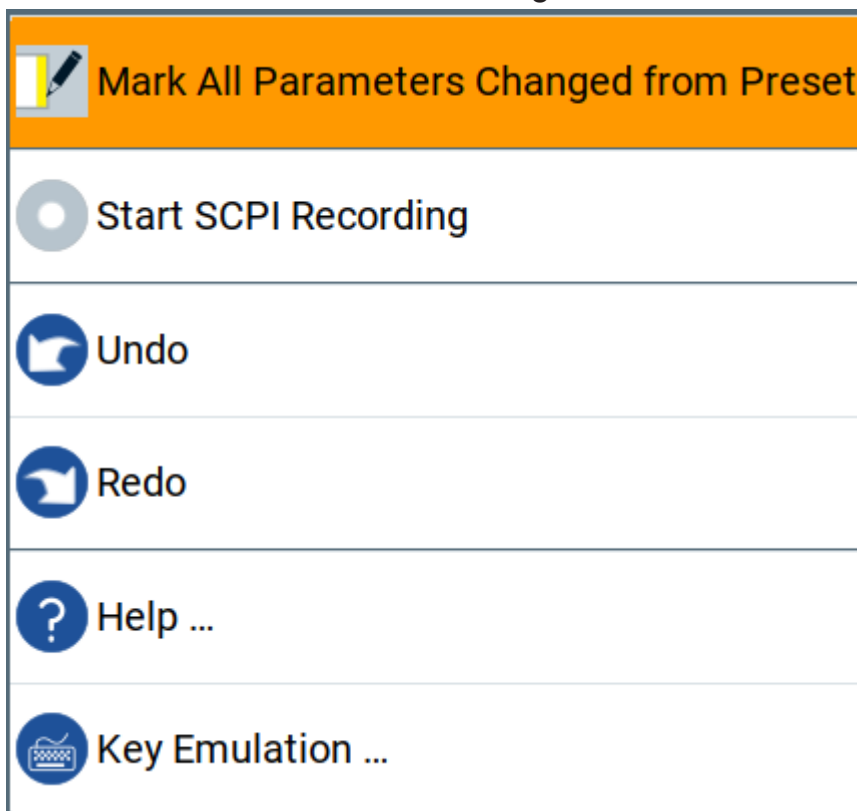


How to display all parameters with values different to their preset values

When you load a file to your instrument, you do not have enough information on the changed settings. In such case, it is useful to visualize all parameters that have been changed from their default state.

Try out the following:

- Touch and hold a spot in the tile diagram for at least 1 second to access the context-sensitive menu.
- Select "Mark All Parameters Changed from Preset".



- All changed parameters are highlighted.

7 Operating the instrument

This chapter provides an overview on basic operating tasks. There are three ways to operate the AREG.

Manual operation

Use the touchscreen, keys and rotary knobs, or an optional mouse and/or keyboard. The principles of manual operation are explained in this section.

Remote control

Create programs to automatize repeating settings, tests, and measurements. The instrument is connected to a computer that runs the program.

This way of operation is described in the user manual, chapter "Network operation and remote control".

Remote operation

For remote monitoring and operation of the instrument, a VNC server is installed on the AREG. You need a LAN connection to the computer, and a VNC client or browser to connect to the instrument.

This way of operation is described in the user manual, chapter "Network operation and remote control".

The following sections show how to operate the instrument manually.

- [Means of manual interaction](#)..... 52
- [Understanding the display information](#)..... 52
- [Accessing the functionality](#)..... 57
- [Entering data](#)..... 58
- [Undo and redo actions](#)..... 60
- [Getting information and help](#)..... 60
- [Remote operation over web UI](#)..... 62

7.1 Means of manual interaction

For the manual interaction with the AREG, you have several methods that you can use as an alternative to perform a task:

- **Touchscreen:**

Touchscreen operation is the most direct way to interact. Almost all control elements and actions on the screen are based on the standard operating system concept. You can tap any user interface element to set parameters in dialog boxes, enter data, scroll within a dialog etc., as if you work with a mouse pointer.

Tapping the screen works like clicking mouse buttons:

- Touch quickly = click: Selects a parameter or provokes an action.
- Touch and hold = right-click: Opens a context-sensitive menu.
- Touch and swipe = drag: Scrolls through the contents of a display element larger than the screen, e.g. a list or a table.

- **Function keys and rotary knob:**

The front panel provides nearly all functions and controls to operate the instrument in the classic way, without touchscreen.

- **Optional mouse and/or keyboard:**

These devices work like known from PCs. The navigation keys on the front panel correspond to the keys on the keyboard.

This manual describes the manual interaction with the instrument via the touchscreen. It mentions the alternative methods using the keys on the instrument or the on-screen keypads if it deviates from the standard operating procedures. The usage of the touchscreen and navigation keys is described in [Chapter 7.3, "Accessing the functionality"](#), on page 57.

Throughout the manual, the term "select" refers to any of the following methods:

- Using a finger on the touchscreen
- Using a mouse pointer in the display
- Using a key on the instrument or on a keyboard

7.2 Understanding the display information

The home screen of the AREG displays all main settings and generator states, divided into main operation areas.

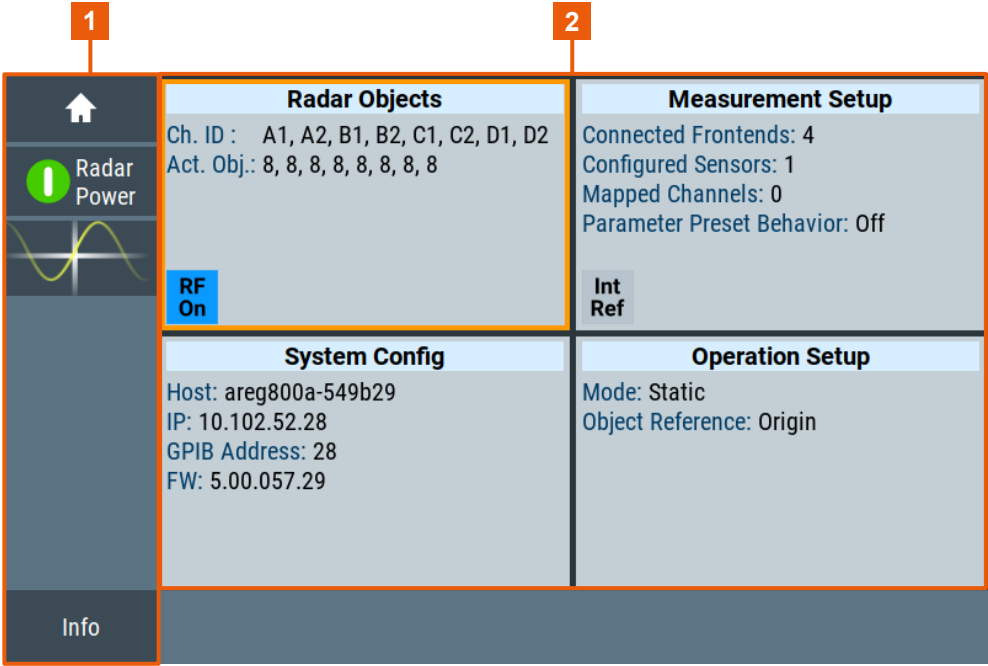


Figure 7-1: Home screen

1 = Taskbar/softkey bar with "Home" and "Info" key
2 = Tile diagram

• Taskbar.....	53
• Tile diagram.....	54
• Additional display characteristics.....	55

7.2.1 Taskbar

The "Taskbar" shows a home and an info button, and assigns a labeled button whenever you open a dialog. If more dialogs are open than the taskbar can display, touch and swipe the taskbar to scroll up and down.

The buttons shown in the following example represent the variants.

Table 7-1: Example of buttons in the taskbar

	"Home" button Returns to the home screen.
	"Busy" indicator Indicates a running process.

Understanding the display information

	"User Menu" Lists parameters that can be defined for quick access.
	Shows the current remote access connections when the instrument is remotely controlled. Tip: An indicator in the "System Config" tile shows the current remote control status.
	R&S NRP power sensors Shows a connected external power sensor. The button displays the current readings when the sensor is active.
	Active dialogs Indicates the dialog name of each active dialog in a separate button.
	"Info" key Provides access to status and error messages. Note: The warning symbol signifies a permanent error message.
	Graphics configuration Opens a dialog to configure the graphical signal monitoring.

7.2.2 Tile diagram

The tile diagram is the main entry to the settings of the AREG.

Tile	Access to:
"Radar Objects"	• RF ON • Configuration • Units • Radar Power • Overview • Graphics
"Measurement Setup"	• Preset Behavior On • Configuration • Channel Mapping • Units • Overview • Reference Frequency • Power Sensors – NRP Power Viewer – NRP Sensor Mapping

Tile	Access to:
"System Config"	• Setup • Info/Notifications • Remote Access • Save/Recall
"Operation Setup"	Opens the operation setup dialog with tabs • Operation Setup • Bandwidth Config • Real-time Control Network • System Control Network

7.2.3 Additional display characteristics

The following section provides a short insight on the indication of the screen in general, and significant elements that you see under specific modes, in dialogs or settings.

- **Appearance of active elements**

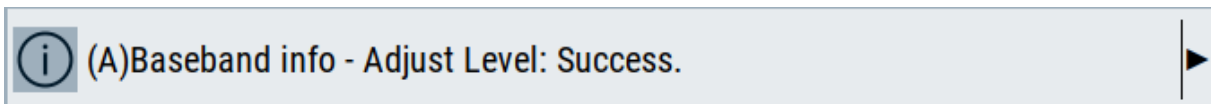
- *Active* elements like On/Off switches, state buttons have a **blue** background.
- *Selected* elements are framed or highlighted **orange**.
- *Inactive* elements are **gray**.

- **On-screen keypads**

Either a numerical or alphanumerical on-screen keypad appears when you enable an entry field, see [Chapter 7.4, "Entering data"](#), on page 58.

- **Info line**

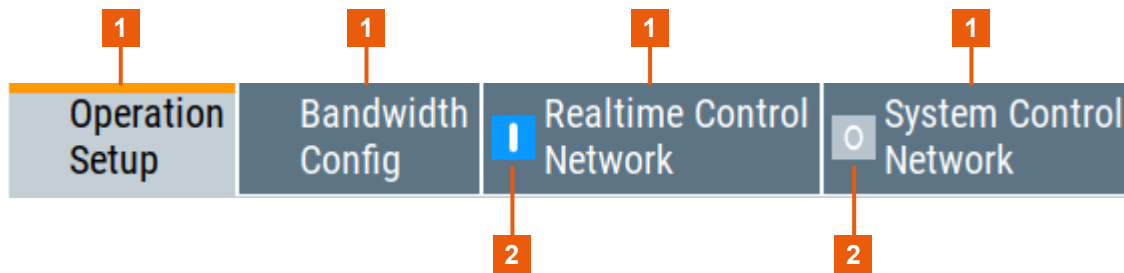
The "Info line" shows brief status information and error messages. It appears when an event generates a message. If selected, the AREG shows information on static errors and the error history.



- **Key parameters indicated in tab labels**

Most dialogs are divided into tabs with logically grouped parameters. The tab label expresses the content and can also contain status indicators.

Understanding the display information



1 = Tab label

2 = Status indicators

- **Scrollbar handle**

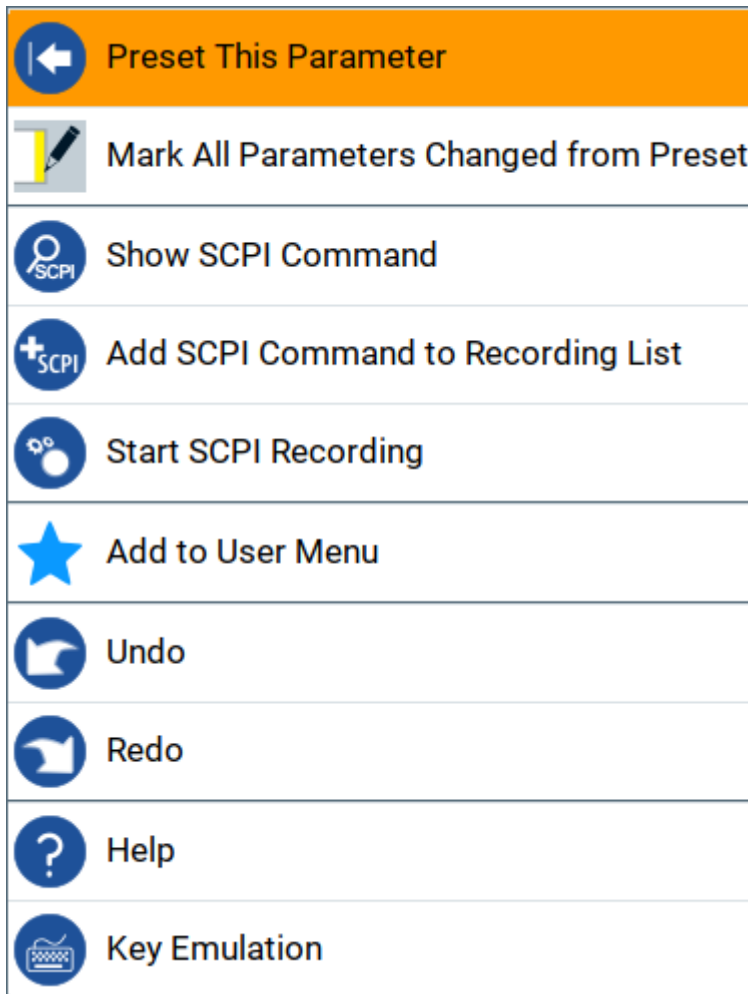
An arrow icon that appears when you touch a scrollbar helps you to scroll in a dialog or list.

- **Progress indicators**

A busy icon indicates a currently running process. If a process takes some time, a progress bar shows the current state.

- **Context-sensitive menus**

Within the entire screen display, including single parameters, you can access context-sensitive menus that provide some additional functions.



7.3 Accessing the functionality

All functionalities are provided in dialog boxes as known from computer programs. You can control the instrument intuitively with the touchscreen. This section provides an overview of the accessing methods.

The instrument's functions and settings can be accessed by selecting one of the following elements:

- System and function keys on the front panel of the instrument
- Taskbar/softkeys on the touchscreen
- Context-sensitive menus for specific elements on the touchscreen, or with the rotary knob (press and hold).
- Elements on the status bar in the touchscreen

- Displayed setting on the touchscreen that means block diagram and all settings available in dialogs.

To open a dialog box

- ▶ Perform one of the following actions:
 - Select the required tile, and then the menu entry.
 - Select the minimized view (button) on the taskbar.

Some of the utility keys access a dedicated dialog, too.

To minimize a dialog box

- ▶ To return to the home screen, select the "Home" button.

To close a dialog box

To close a dialog box, you have the same controls as you know from computers or devices with touchscreen.

- ▶ Perform one of the following actions:
 - Select the "Close" icon in the upper right corner.
 - Select the [Esc] key on the front panel.
 - Drag and drop a minimized dialog from the taskbar to the tile diagram.

To select an option in a dialog box

- ▶ Select the required option.

7.4 Entering data

Some parameters have their own key on the front panel. For data input in dialog boxes, the instrument provides on-screen keypads for entering numeric and alphanumeric values. You can always set the parameters via the touchscreen, the front panel or an external keyboard.

To enter numeric values with the on-screen keypad

For numeric settings, the instrument displays the numeric keypad. The units specified correspond to the units of the parameter.

1. Enter the numeric value.

Tip: For a quick entry, you can enter a value in an exponential representation, for example *1e7* for *10000000*.

2. Tap the unit button to complete the entry.

The unit is added to the entry.

Tip: For a quick unit change, you can enter shortcuts, for example for a frequency value *1e8h* for *100 MHz*.

For an overview of shortcuts supported by the AREG, see chapter "Unit Shortcuts" in the appendix of the user manual.

3. If the parameter does not require a unit, confirm the entered value by pressing "Enter".

To enter numeric values with the front panel controls

1. Change the currently used parameter value by using the rotary knob or the [Up/Down] keys.
2. If the parameter does not require a unit, confirm the entered value by pressing the [Enter] key or any of the unit keys.
The instrument highlights the editing line to confirm the entry.

If you edit numeric data in tables, enable edit mode first. Press [Enter] or press the rotary knob.

To enter alphanumeric values

If a field requires alphanumeric input, you can use the on-screen keyboard to enter letters and characters including special characters.

To complete an entry

- Press the [Enter] key or the rotary knob.

To correct an entry

1. Using the arrow keys, move the cursor to the right of the entry you want to delete.
2. Press the [Backspace] key.
3. Deletes the entry to the left of the cursor.
4. Enter your correction.

To abort an entry

- ▶ Press the [Esc] key.
The dialog box closes without changing the settings.

7.5 Undo and redo actions

Accessed via the context-sensitive menus, "Undo" allows you to restore one or more actions on the instrument. Depending on the available memory, the "Undo" steps can restore all actions.



"Redo" restores a previously undone action.

7.6 Getting information and help

In some dialog boxes, graphics are included to explain the way a setting works.

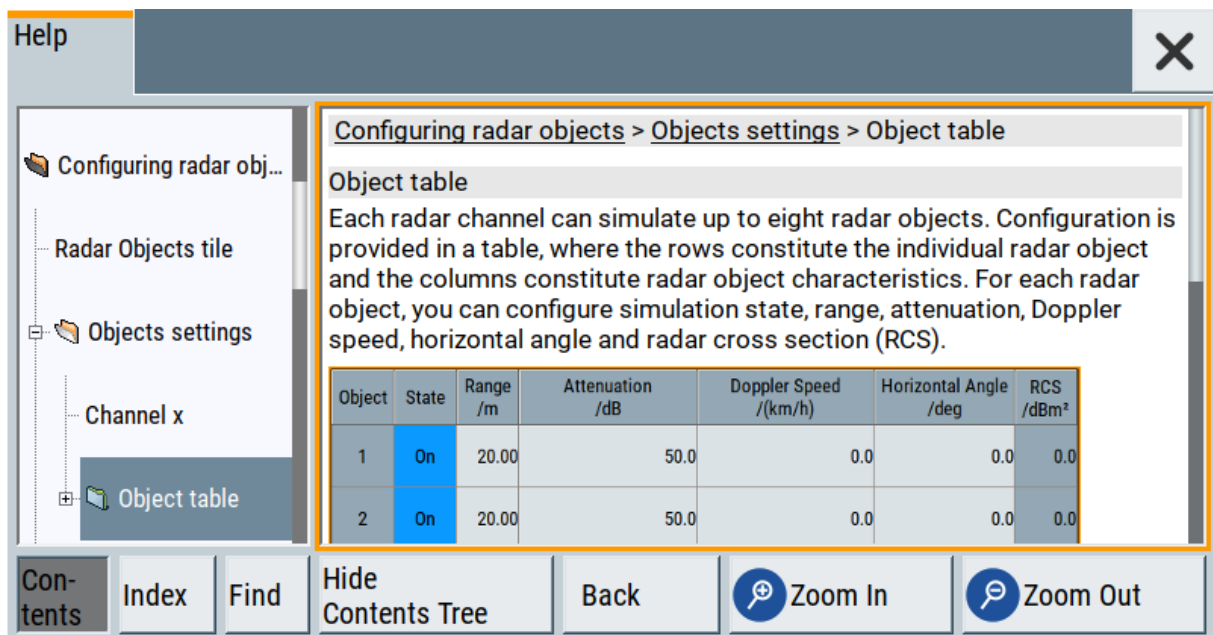
For further information, you can use the following sources:

- Tooltips give the value range of the parameter.
- The context help provides a functional description on a setting.
- The general help explains a dialog box, provides instructions, and general information.

To display context help

- ▶ To access a help topic, perform one of the following:
 - a) Tap and hold the parameter for which you need information.
 - b) In the opening context menu, tap "Help".
 - c) Tap the parameter.
 - d) Press the [Help] key.

The "Help" dialog opens. You can browse the help for further information.



Contents of the help dialog box

The help dialog box covers two main areas:

- "Contents" - contains a table of help contents
- "Topic" - contains a specific help topic

The help system also provides an "Index" and a "Find" area, and "Zoom" functions that are accessed via the corresponding buttons.

To open general help

- Press the yellow [Help] key on the front panel.

If a dialog box is opened, the help topic for the current tab is shown. Otherwise the "Contents" page appears.

Navigating in the table of contents and in the help topics

1. To move through the displayed content entries, tap on an entry. Entries with a plus sign contain further entries.
2. To display a help topic, tap on the topic name.
3. To follow a cross-reference, tap on the link text.
4. To return to the previous page, select "Back".
This function scrolls back all steps that you have performed before.
5. Use the "scroll bars" to shift the visible section of content shown.

Remote operation over web UI

6. To maximize the "Topics" area, tap the "Hide Contents Tree" button to hide the contents tree.

Using the index

1. Select the "Index" button.
2. Enter the first characters of the topic that you are interested in.
The entries starting with these characters are displayed.
3. Tap on the index entry.
The corresponding help topic is displayed.

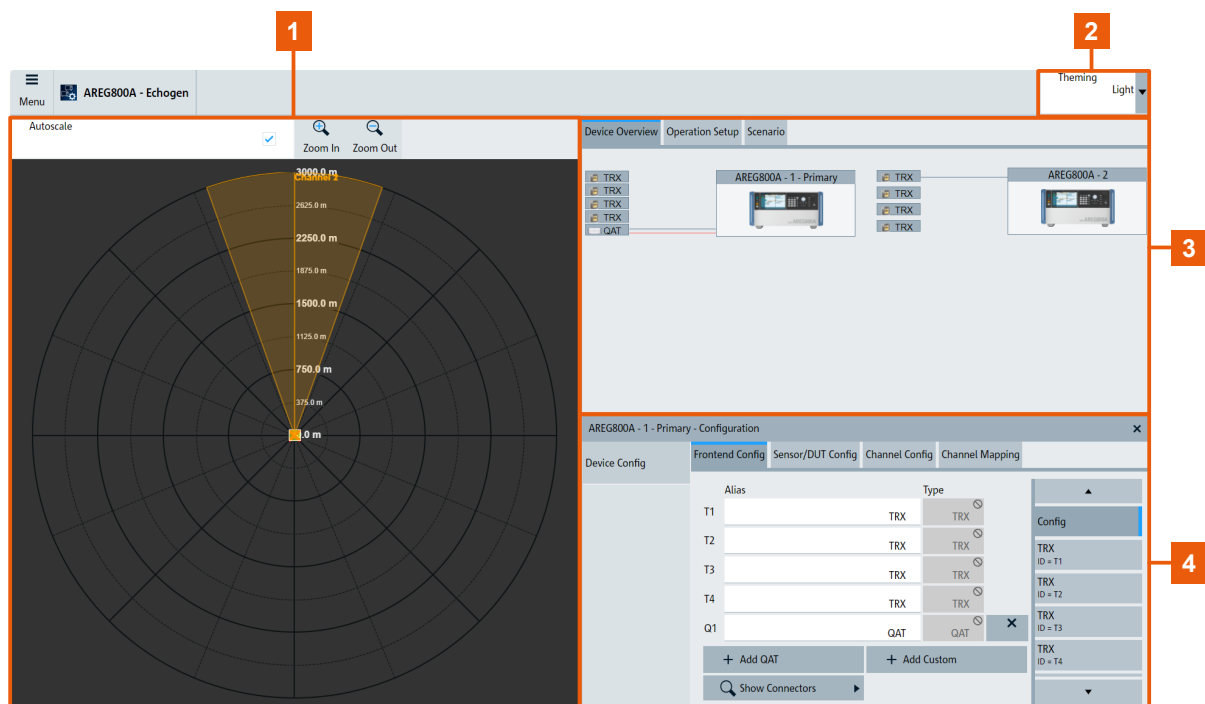
7.7 Remote operation over web UI

You can configure the AREG, for example in a multi-instrument test setup, over a web user interface (web UI). All instruments require connection to the same LAN.

See also the AREG user manual, chapter "Test setup for multiple instruments".

Access:

- In the address field of the browser, enter *<ipaddress>/webui*.



- 1 = [Overview area](#)
- 2 = [Theming](#)
- 3 = [Setup area](#)
- 4 = [Configuration area](#)

The web UI allows you to define your operation setup and configure the connected instruments.

Settings:

Overview area	63
Theming	63
Setup area	63
Configuration area	63

Overview area

Displays an overview of all objects within the radar object simulation scenario.

Theming

Selects the theming of the web UI interface. You can select light themings or dark themings.

Setup area

In this area, you can define your operation setup.

"Device Overview"

Displays the instruments in the test setup. Tap on the instrument to select the respective AREG for configuration in the configuration area.

"Operation Setup"

Define the operation setup settings for all connected instruments.

"Scenario"

Requires "Operation Setup > Mode > Dynamic".
Select a scenario file stored on your instrument and play the scenario.

Configuration area

You can configure the selected AREG.

8 Contacting customer support

Technical support – where and when you need it

For quick, expert help with any Rohde & Schwarz product, contact our customer support center. A team of highly qualified engineers provides support and works with you to find a solution to your query on any aspect of the operation, programming or applications of Rohde & Schwarz products.

Contact information

Contact our customer support center at www.rohde-schwarz.com/support, or follow this QR code:



Figure 8-1: QR code to the Rohde & Schwarz support page

Index

A

Active elements	55
Alphanumeric values	
Entering	59
Application cards	15
Application notes	15

B

Bench top, placing	18
Brochures	14

C

Calibration certificate	14
Carrying the instrument	17
Checking the instrument	17
Connecting	
Keyboard	23
LAN	22
Memory stick	23
Mouse	23
Power	21
To Digital IO DD	27
To Ref In/Ref Out	27
To RF coaxial connectors	24
USB devices	23
Connecting to RF coaxial connectors	
To connect to pluggable connectors ...	25
To connect to screwable connectors ...	24
To prepare for connecting	24
Connector	
AC power supply	43
GPIB	42
IEC/IEEE	42
LAN	43
Ref In	42
Ref Out	42
USER	42
Context-sensitive menu	55
Customer support	64

D

Data entry	58
Data sheets	14
Dialog boxes	57

Display

Active elements	55
Context-sensitive menu	55
Info line	55
Information	52
On-screen keypad	55
Tab labels	55
Taskbar	53
Tile diagram	54
Documentation overview	13

G

Getting started	13
-----------------------	----

H

Help	13, 60
------------	--------

I

Info line	55
Input connector	42
Instrument	
Carrying	17
Checking	17
Lifting	17
Operating site	18
Unpacking	17
Instrument help	13
Instrument security procedures	14

K

Key

Arrow	38
Backspace	37
Down	38
Enter	37
Esc	37
Freq	36
Help	35
Home	38
Left	38
Level	36
Local	35
Mod On/Off	36
Next window	38
On/Off	38
On/Standby	35
Preset	35

RF On/Off	36
Right	38
Setup	35
Undo	38
Up	38
User	38
Key features	16
Keyboard	
On-screen	58
Usage	52
Keypad	
On-screen	58
L	
LAN	
Connecting	22
Environment	22
Lifting the instrument	17
Loading	
Trying out	49
M	
Manual interaction	52
Mounting, in a rack	19
Mouse	
Usage	52
N	
Navigation	
Keys	38
Network	
Environment	22
Numeric data entry	58
Numeric values	
Entering	58
O	
On-screen keypad	55
Open source acknowledgment (OSA)	15
Operating site	
Choosing	18
Setting up the instrument	18
Operation	
Concepts	51
Manual	52
Output connector	42
P	
Placing, on a bench top	18

Power	
Connecting the instrument	21
Power on	
Key	35
Power supply	
Connector	43
Preparing for use	17
Connecting R&S QAT100	26

R

Rack, mounting	19
Release notes	15
Remote control	51
Remote Desktop	51
Remote operation	51
Rotary knob	37

S

Safety instructions	7, 14
Security procedures	14
Service manual	13
Softkey bar	
See Taskbar	53
Specifications	14
Standby	
Key	35
Switching	
On or off	31

T

Tab labels	55
Text entry	58
Tile diagram	
Display	54
Tooltips	
Show	60
Touchscreen	
Compared with mouse	52
Overview	34
Usage	52

U

Unpacking the instrument	17
User manual	13
Utility keys	
Overview	35

V

Videos	15
VNC	51

W

Web interface	51
White papers	15